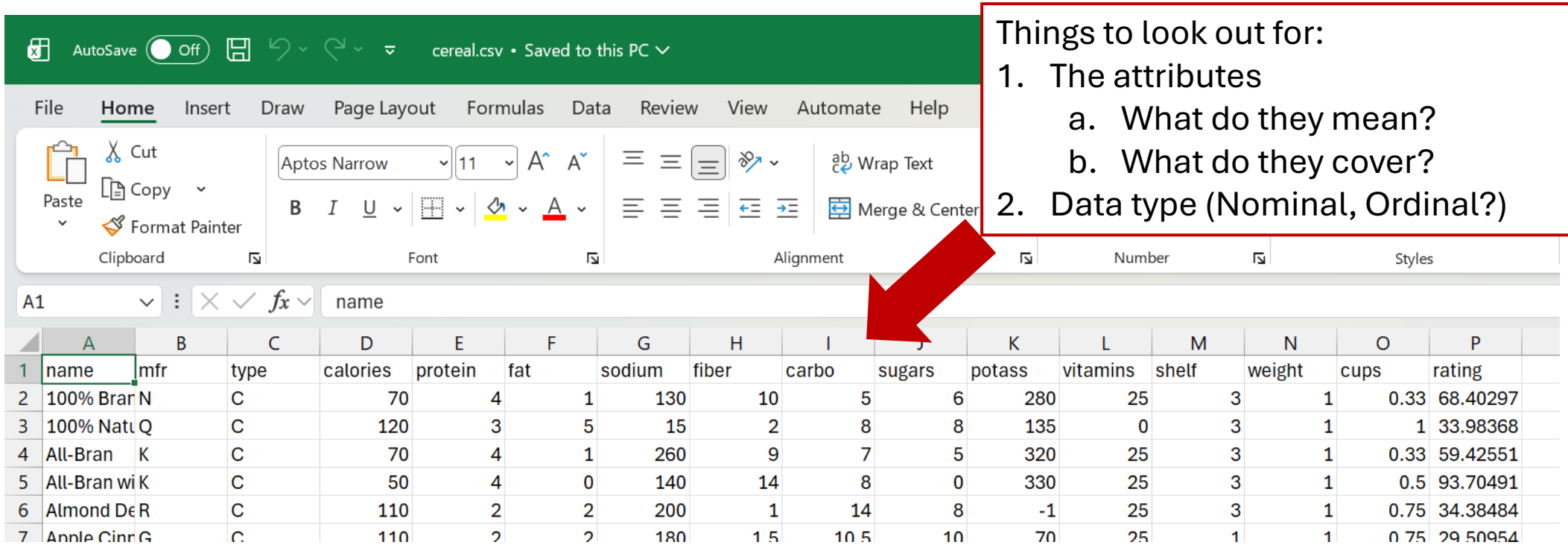
Lab 1:
Tools - DataWrapper, Inzight, RAWGraphs

Spring COMP4462

Before using the tools…

- Get familiar with the dataset (cereal.csv)

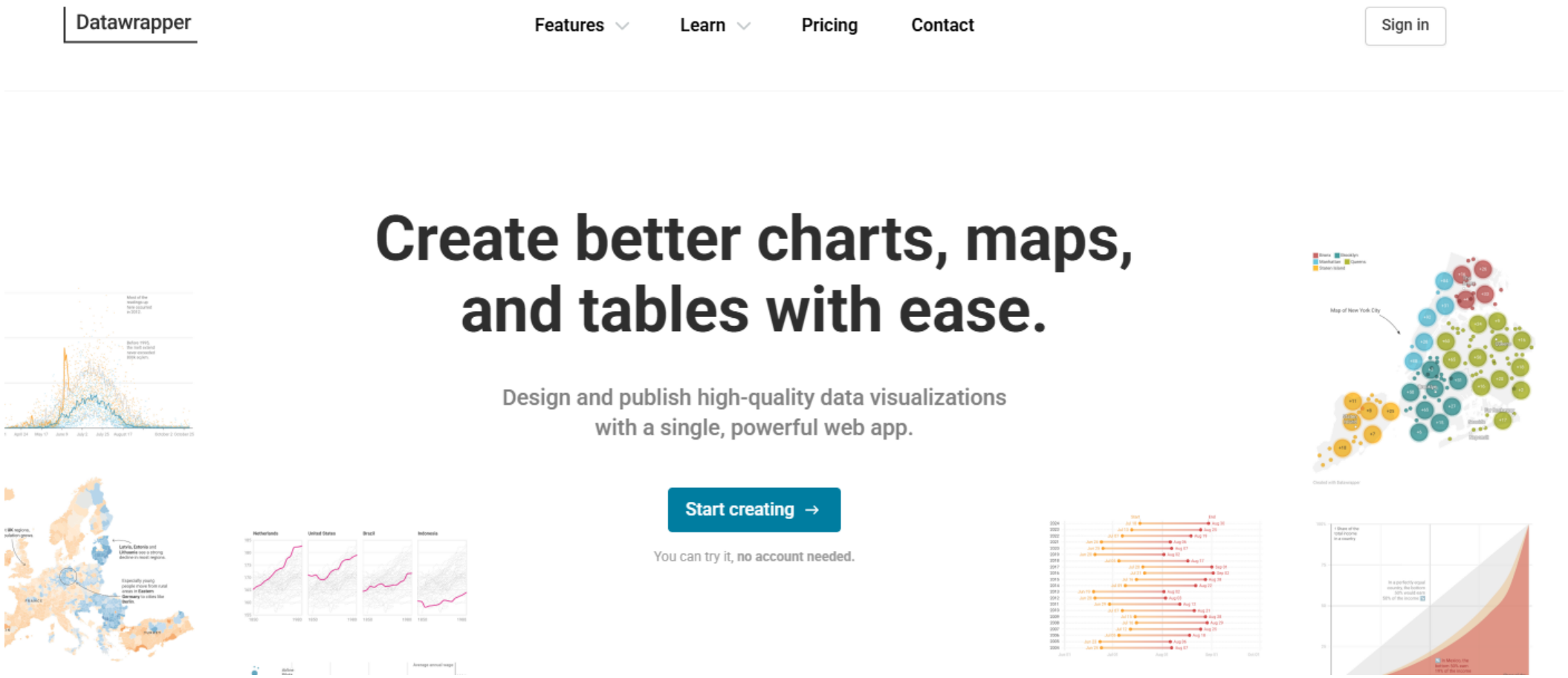


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	name	mfr	type	calories	protein	fat	sodium	fiber	carbo	sugars	potass	vitamins	shelf	weight	cups	rating
2	100% Bran	N	C	70	4	1	130	10	5	6	280	25	3	1	0.33	68.40297
3	100% Natu	Q	C	120	3	5	15	2	8	8	135	0	3	1	1	33.98368
4	All-Bran	K	C	70	4	1	260	9	7	5	320	25	3	1	0.33	59.42551
5	All-Bran wi	K	C	50	4	0	140	14	8	0	330	25	3	1	0.5	93.70491
6	Almond De	R	C	110	2	2	200	1	14	8	-1	25	3	1	0.75	34.38484
7	Apple Cinr	G	C	110	2	2	180	1.5	10.5	10	70	25	1	1	0.75	29.50954

Things to look out for:

1. The attributes
   a. What do they mean?
   b. What do they cover?
2. Data type (Nominal, Ordinal?)

Tool 1: DataWrapper



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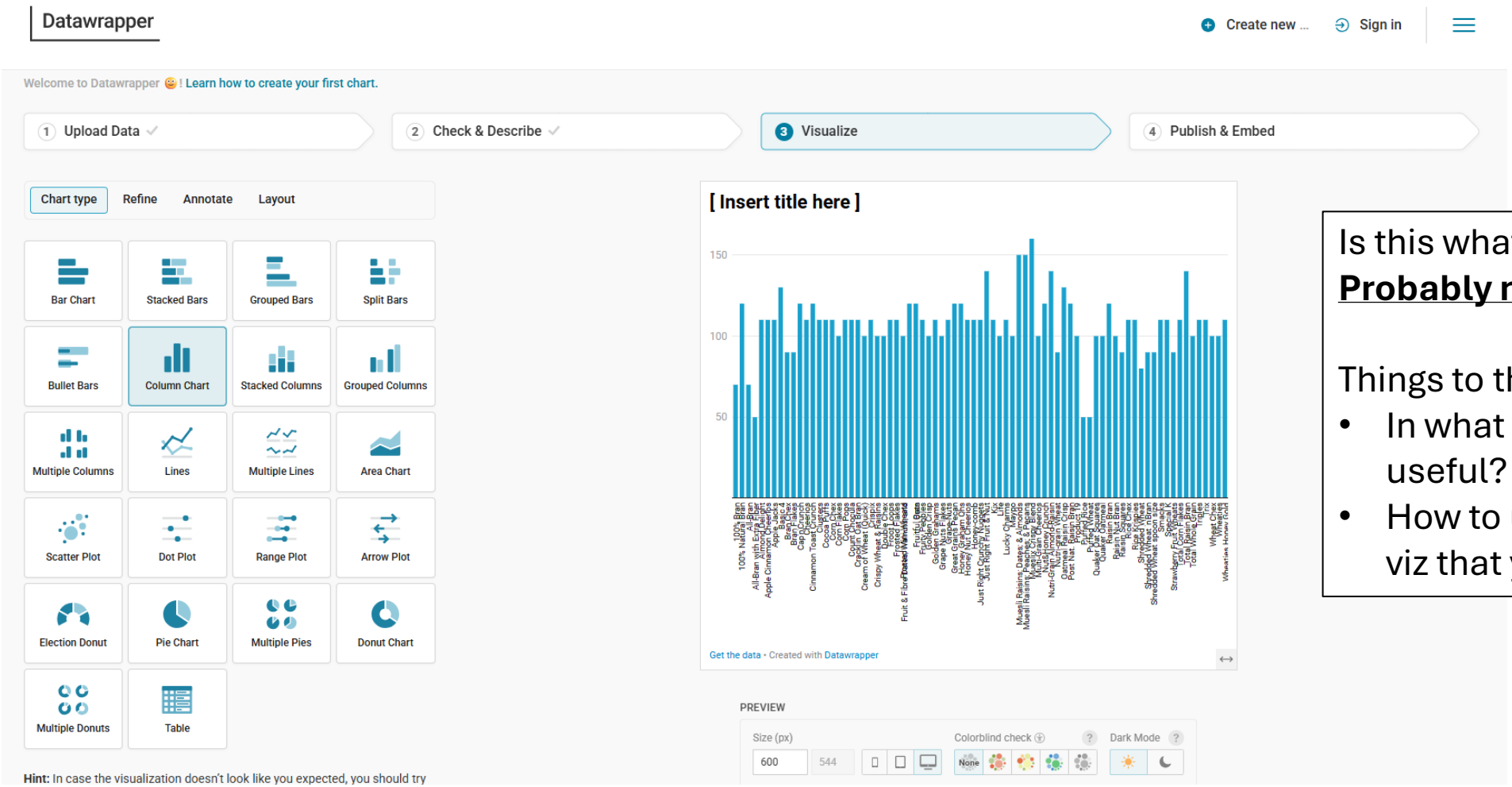


Is this what you want?
Probably not!

Things to think about:

- In what ways is this useful?
- How to use this to create viz that you want?

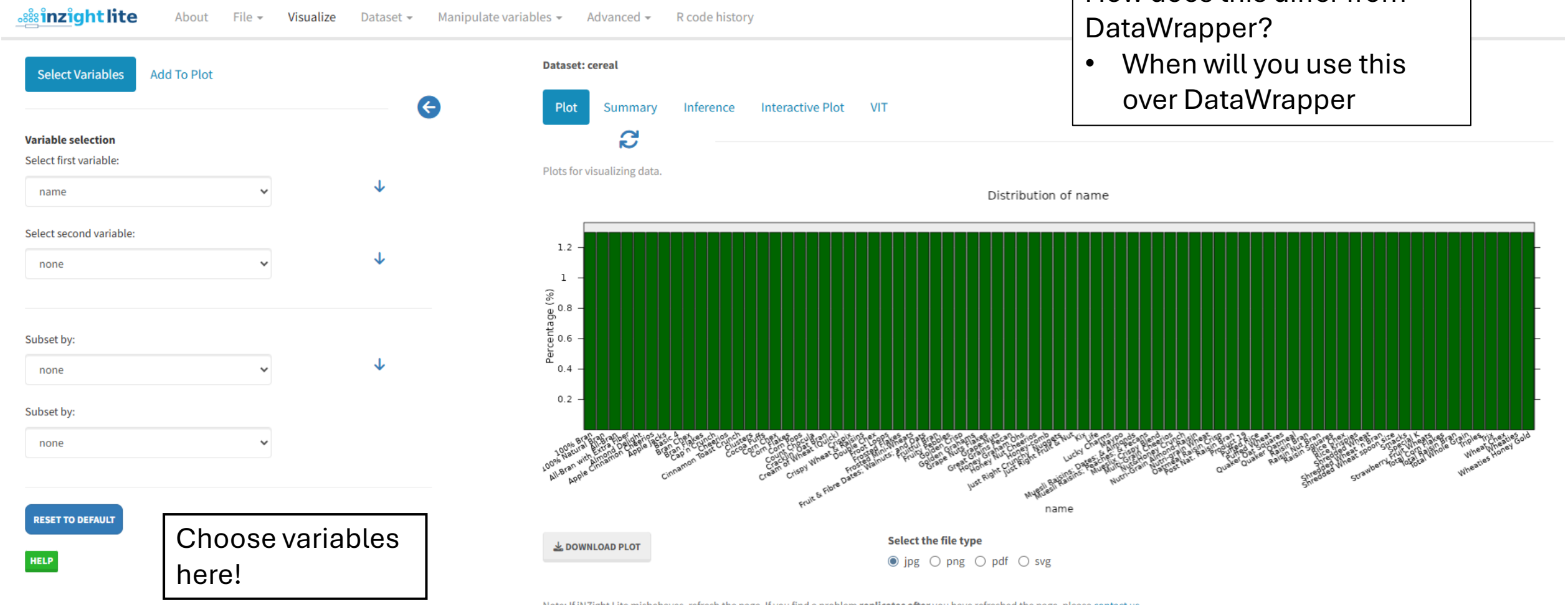
Tool 2: Inzight

Either one is ok

Tool 2: Inzight

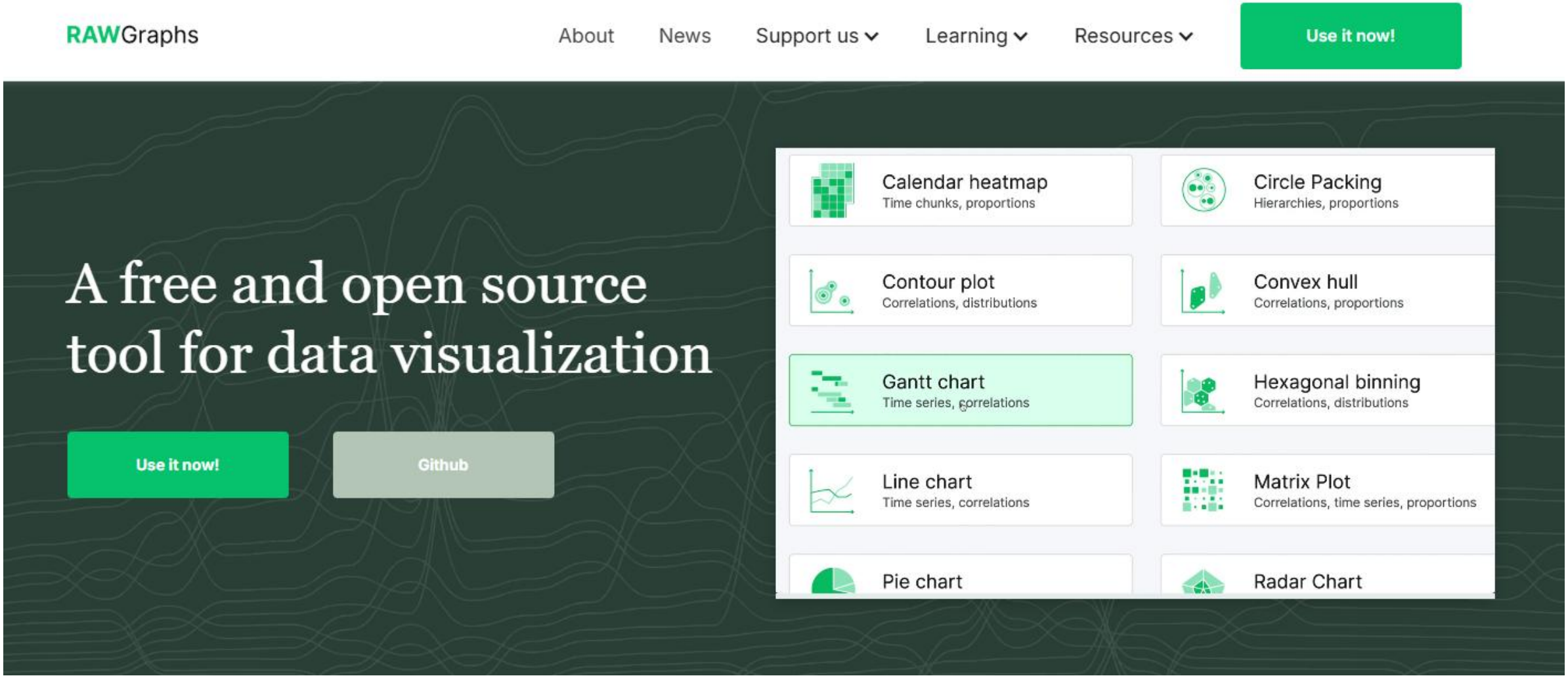
Think about:
How does this differ from DataWrapper?

- When will you use this over DataWrapper

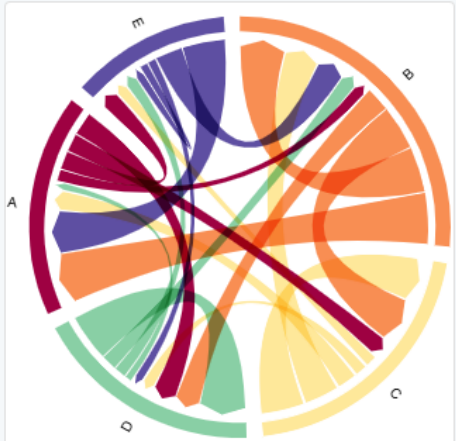


Choose variables here!

Tool 3: RAWGraphs



Tool 3: RAWGraphs



Chord Diagram

It shows relationships among nodes. Nodes size represent the sum of incoming and outgoing links. Relationships are drawn as arcs whose width represent their values.

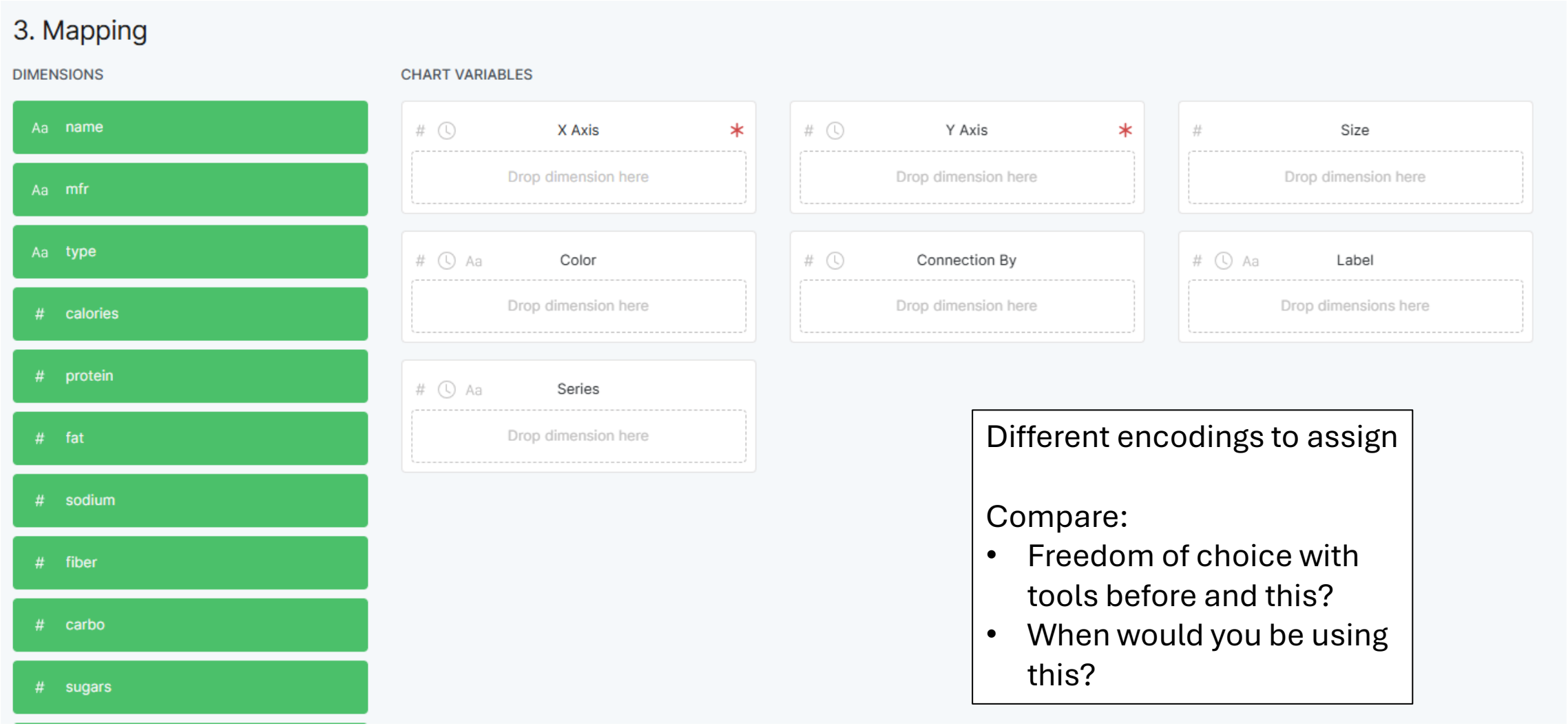
Code Tutorial

- Alluvial Diagram — Correlations, proportions
- Arc Diagram — Networks
- Bar chart — Correlations
- Multi-set bar chart — Correlations, proportions
- Stacked bar chart — Correlations, proportions
- Beeswarm plot — Distributions, time series, proportions
- Box plot — Distributions
- Bubble chart — Correlations, proportions
- Bumpchart — Time series, correlations, proportions
- Calendar heatmap — Time chunks, proportions
- Chord Diagram — Networks
- Circle Packing — Hierarchies, proportions
- Circular dendrogram — Hierarchies, proportions
- Contour plot — Correlations, distributions
- Convex hull — Correlations, proportions
- Linear dendrogram — Hierarchies, proportions
- Gantt chart — Time series, correlations
- Hexagonal binning — Correlations, distributions
- Horizon graph — Time series, correlations
- Line chart — Time series, correlations
- Matrix Plot — Correlations, time series, proportions
- Parallel coordinates — Correlations, distributions
- Pie chart — Proportions
- Radar Chart — Correlations
- Sankey Diagram
- Slope chart
- Streamgraph (area chart)

Different chart types to explore!

- What are the chart types for tools before?

Tool 3: RAWGraphs



Different encodings to assign

Compare:

- Freedom of choice with tools before and this?
- When would you be using this?

Lab Task

- On Canvas, under https://canvas.ust.hk/courses/67750/assignments/412328

Tasks

1. [6 points] Import the cereal dataset into each of the above systems and then try to create charts for each of the following questions:
   1. Do high-fiber cereals tend to have higher ratings?
   2. I want to know the average calories and sugars by brand.
   3. Which cereals are top-5 and bottom-5 by rating?
   4. (Subjective) Which is the "best" cereal that you would want to have and recommend?

Choose a tool to try and create charts to answer per question

2. [4 points] Generally critique the systems. What are their strengths and weaknesses? How do they compare? What kinds of tasks is each best suited for? It's OK to describe your experience actually learning and using the systems, but try not to focus too much on small details and nuances of the UIs. Instead, focus more on the system's utility and their ability to assist data analysis.

Tips

- Sketch out ideas!
- Compare the difference in the tools, what difficulties do you face when using it?
  - In terms of managing data?
  - In creating aggregated/ derived fields?
  - Is it intuitive?
  - Can it do all the things I want?
  - Others…

Reminder

- Lab task deadline: Feb 20 23:59
- **Group project!**
  - Form a group, if you have not remember to use Piazza to look for teammates!
  - Start thinking of a topic if you have a group
- Take attendance if you have not yet!

Questions?